Week 6 – Design Studio

Design the airline's disruption assistant

The system. A storm closes the airline's hub for the evening. Every passenger on a cancelled flight opens the app and asks what they can do. The assistant answers from the airline's own documents, for this passenger and this flight, as things stand right now, with the rule or waiver it relied on quoted. The passenger rebooks, claims, or is handed to an agent.

the fare rules	2,000 documents; change monthly, through review
the waivers	published by operations during a disruption: routes, dates, what the passenger may do, until when. 5 to 40 a day; changed or withdrawn within the hour
the flight status	1,500 flights; status, gate, new time and rebooking options change by the minute
the booking	this passenger's flight, fare class, what they have already been offered; nobody else's
questions	20,000 on an ordinary day; 300,000 on a storm evening, **150 a second** at the peak
the answer	within **5 s**, the waiver quoted
the vendor	**50 generations a second** contracted
unanswerable	about a third: compensation the rules do not give, a promise an agent made by phone
ground truth	a rebooking in the session; a claim decided in days; a complaint in weeks; mostly nothing

Task.

1. **Architecture.**
   - Draw the whole system: the services, the jobs, the stores, and the flow between them
   - Show how a fare rule, a waiver, a flight-status update and the booking each reach an answer, and how old each can be when the passenger reads it
2. **Measurement.**
   - Per stage, before any passenger sees it: what is measured, against what
   - The unanswerable slice: where its questions come from, including last week's answerable questions whose waiver has gone; how many the floor stops before generation, and how many

the model answers anyway

- What decides a release, and one guardrail that can stop a release whose primary metric rose

Your design should handle:

1. Each source at its rate: indexed, summarized ahead, or looked up at the request; how a change reaches the next answer; the as-of the answer carries
2. The booking in the context, never in the shared index
3. A waiver as a document: how it is chunked, what rides as metadata, what happens the minute it expires
4. Ten thousand passengers asking the same question in the same hour: the cache, its key, and what makes an entry wrong
5. Anything that commits the airline to money is quoted verbatim with the waiver's id: what checks that, on every answer
6. When the assistant does not answer: what the passenger sees, what the agent receives

(Bonus) 150 questions a second against 50 generations a second: what the passenger still gets, and what the system never does.

One question. At 19:40 the assistant told a passenger their hotel was covered, and quoted the waiver. The waiver was withdrawn at 19:30. What does the log hold, which component let it through, and which should have stopped it?